Project #2: Cache & Memory Performance Profiling

Modern CPUs use a deep hierarchy of L1/L2/L3 caches and DRAM. Each level differs in latency, bandwidth, capacity, and queuing behavior. This project measures the hierarchy and explains how locality, concurrency, cache misses, and TLB misses determine performance.

Learning Goals

- Measure zero-queue latency for L1, L2, L3, and DRAM where feasible.
- Measure DRAM bandwidth under different access granularities, strides, and read/write mixes.
- Reveal the throughput-latency tradeoff as memory access intensity grows.
- Correlate cache-miss and TLB-miss behavior with kernel performance.
- Use Little's Law and AMAT-style reasoning to interpret results.

Tools

- Intel Memory Latency Checker or an equivalent memory benchmark for latency/bandwidth sweeps.
- perf or equivalent counters for cache/TLB behavior, cycles, instructions, and memory events.
- Simple loop kernels, such as z[i] = x[i] * y[i] or y[i] = a*x[i] + y[i], used to correlate cache/TLB misses with runtime.

Experimental Knobs

- Access pattern: sequential vs. random; strides near 64B, 256B, and 1024B.
- Read/write ratio: 100% read, 100% write, 70/30, and 50/50 where supported.
- Access intensity: outstanding requests or threads, including loaded-latency sweeps.
- Working-set size: range from L1-resident to DRAM-resident.
- Page locality: 4KB pages vs. huge pages where available.

Required Experiments and Plots

- Zero-queue baselines: per-level latency table with units and CPU-frequency conversion where useful.
- Pattern and granularity sweep: latency and bandwidth across sequential/random and stride/granularity settings.
- Read/write mix sweep: compare throughput and latency under the required mixes.
- Intensity sweep: plot throughput vs. latency, identify the knee, and relate it to Little's Law.
- Working-set sweep: annotate L1/L2/LLC/DRAM transition regions.
- Cache-miss impact: vary footprint or access pattern and correlate perf counters with runtime.
- TLB-miss impact: vary page locality and evaluate huge pages where possible.
- Show one benchmark pitfall, such as accidentally measuring cache reuse instead of DRAM, failing to pin threads, or allowing frequency scaling to distort results.

Deliverables

- Scripts/commands, raw data, processed data, and plotting code.
- System configuration: CPU, memory, OS/kernel, tool versions, governor/frequency policy, SMT state, pinning, NUMA placement if relevant.
- Plots/tables with units, repeated trials, and error bars when feasible.
- Analysis tied to latency, bandwidth, concurrency, cache/TLB counters, AMAT, and Little's Law.

Grading Rubric

Total: 185 points

Category	Points	Description
Zero-queue baselines	25	Methodologically sound latency measurements with clear units and caveats.
Pattern/granularity/RW sweeps	35	Complete and well-controlled bandwidth/latency comparisons.
Throughput-latency knee	35	Clear knee identification, diminishing-return discussion, and Little's Law connection.